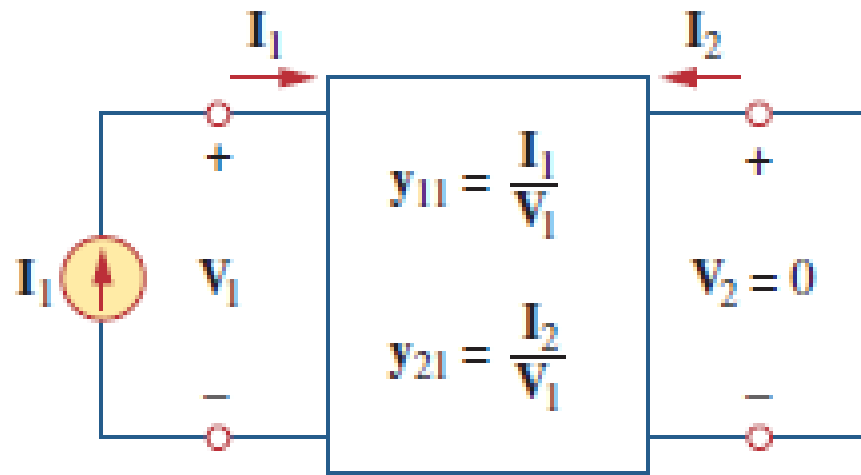


Module - 5

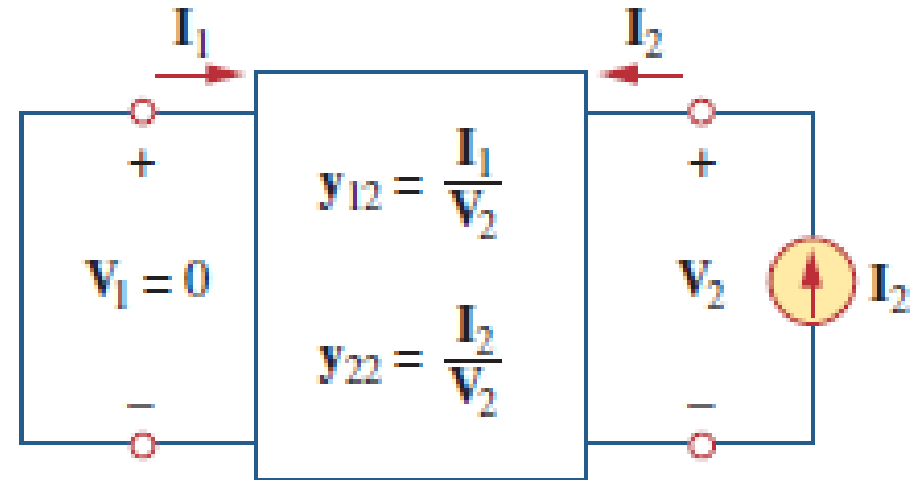
Y - Parameters

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SENSE

Admittance Parameters



(a)



Y terms are called admittance parameters or simply y parameters or short circuit parameters with the units of mho.

$$\mathbf{I}_1 = y_{11}\mathbf{V}_1 + y_{12}\mathbf{V}_2$$

$$\mathbf{I}_2 = y_{21}\mathbf{V}_1 + y_{22}\mathbf{V}_2$$

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = [\mathbf{y}] \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

$$y_{11} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_1} \right|_{\mathbf{V}_2=0}, \quad y_{12} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_2} \right|_{\mathbf{V}_1=0}$$

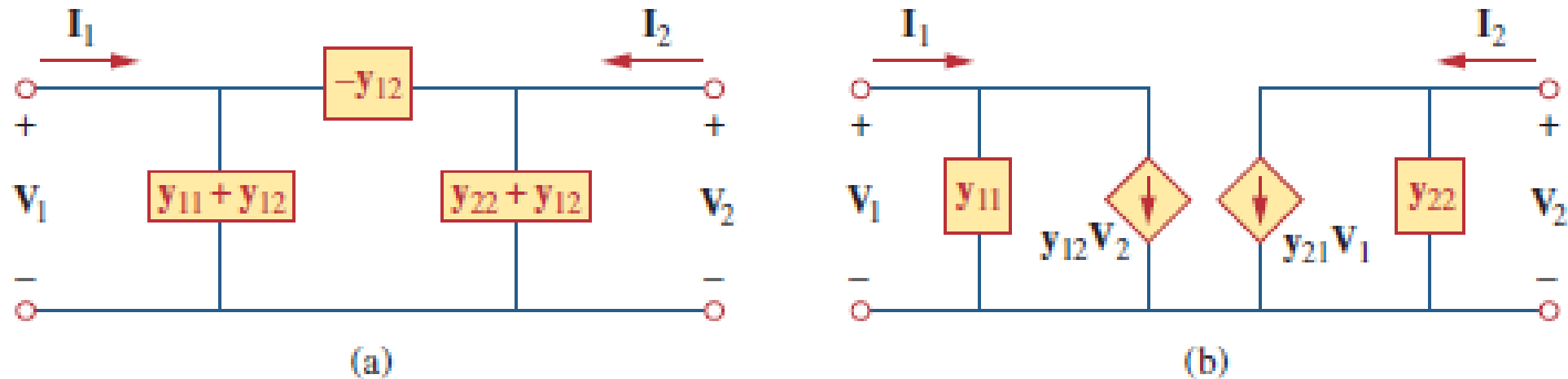
$$y_{21} = \left. \frac{\mathbf{I}_2}{\mathbf{V}_1} \right|_{\mathbf{V}_2=0}, \quad y_{22} = \left. \frac{\mathbf{I}_2}{\mathbf{V}_2} \right|_{\mathbf{V}_1=0}$$

y_{11} = Short-circuit input admittance

y_{12} = Short-circuit transfer admittance from port 2 to port 1

y_{21} = Short-circuit transfer admittance from port 1 to port 2

y_{22} = Short-circuit output admittance



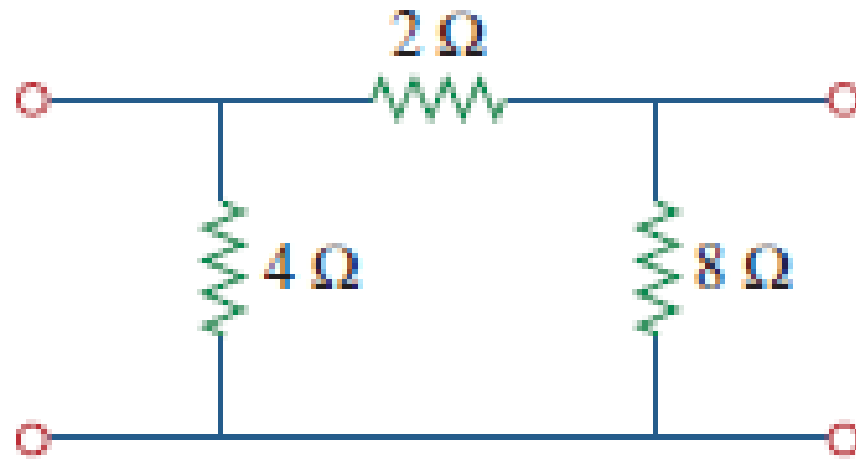
(a) Π -equivalent circuit (for reciprocal case only), (b) general equivalent circuit.

For a two-port network that is linear and has no dependent sources, the transfer admittances are equal ($y_{12} = y_{21}$). This can be proved in the same way as for the z parameters. A reciprocal network ($y_{12} = y_{21}$) can be modeled by the Π -equivalent circuit

If the network is not reciprocal, a more general equivalent network

Numerical -1

Obtain the y parameters for the Π network shown



$$V_1 = I_1(4 \parallel 2) = \frac{4}{3}I_1, \quad y_{11} = \frac{I_1}{V_1} = \frac{I_1}{\frac{4}{3}I_1} = 0.75 \text{ S}$$

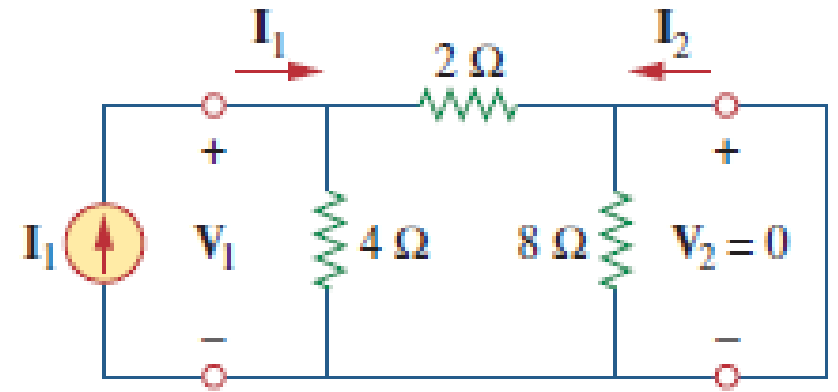
By current division,

$$-I_2 = \frac{4}{4+2}I_1 = \frac{2}{3}I_1, \quad y_{21} = \frac{I_2}{V_1} = \frac{-\frac{2}{3}I_1}{\frac{4}{3}I_1} = -0.5 \text{ S}$$

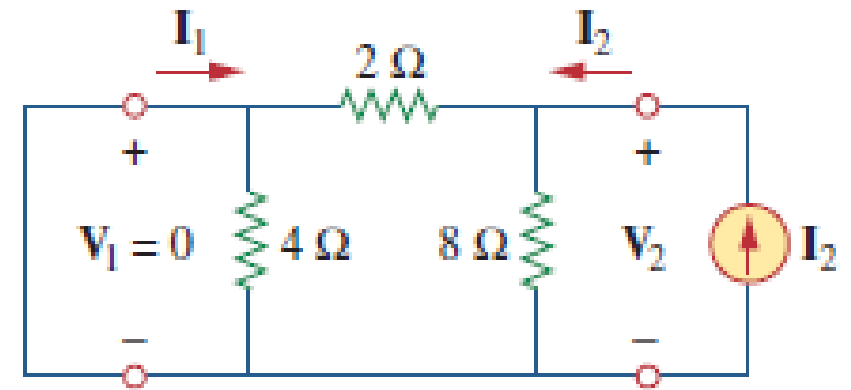
$$V_2 = I_2(8 \parallel 2) = \frac{8}{5}I_2, \quad y_{22} = \frac{I_2}{V_2} = \frac{I_2}{\frac{8}{5}I_2} = \frac{5}{8} = 0.625 \text{ S}$$

By current division,

$$-I_1 = \frac{8}{8+2}I_2 = \frac{4}{5}I_2, \quad y_{12} = \frac{I_1}{V_2} = \frac{-\frac{4}{5}I_2}{\frac{8}{5}I_2} = -0.5 \text{ S}$$



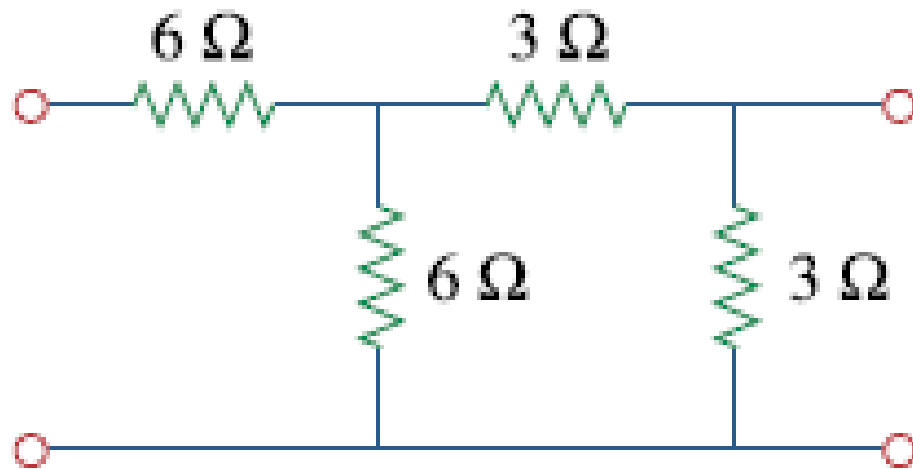
(a)



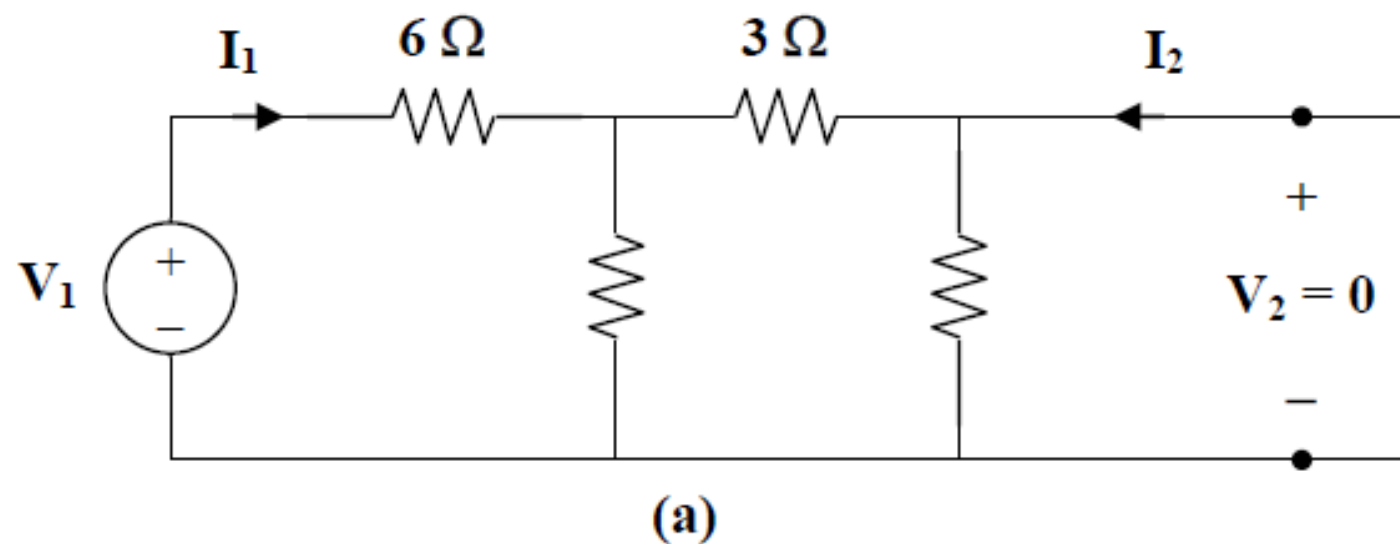
(b)

Numerical-2

Calculate the y parameters for the two-port



To get y_{11} and y_{21} , consider the circuit in Fig.(a).



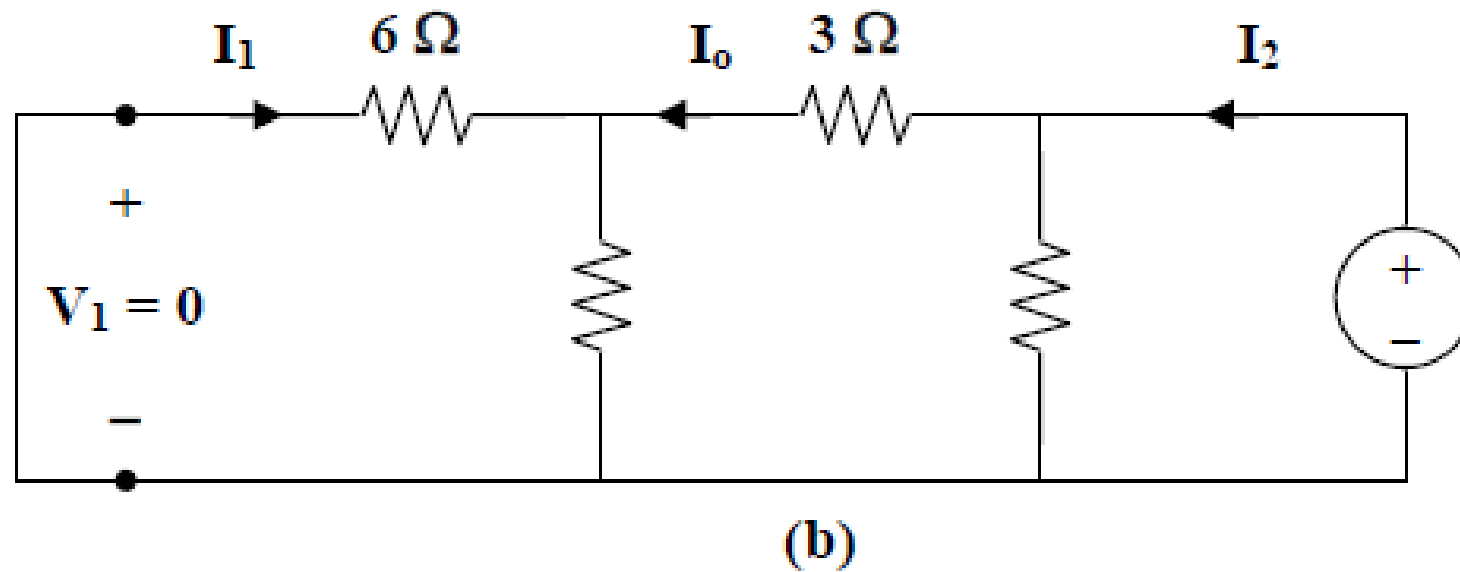
$$V_1 = (6 + 6 \parallel 3) I_1 = 8 I_1$$

$$y_{11} = \frac{I_1}{V_1} = \frac{1}{8}$$

$$I_2 = \frac{-6}{6+3} I_1 = \frac{-2}{3} \frac{V_1}{8} = \frac{-V_1}{12}$$

$$y_{21} = \frac{I_2}{V_1} = \frac{-1}{12}$$

To get y_{22} and y_{12} , consider the circuit in Fig.(b).



$$y_{22} = \frac{I_2}{V_2} = \frac{1}{3 \parallel (3 + 6 \parallel 6)} = \frac{1}{3 \parallel 6} = \frac{1}{2}$$

$$I_1 = \frac{-I_o}{2},$$

$$I_o = \frac{3}{3 + 6} I_2 = \frac{1}{3} I_2$$

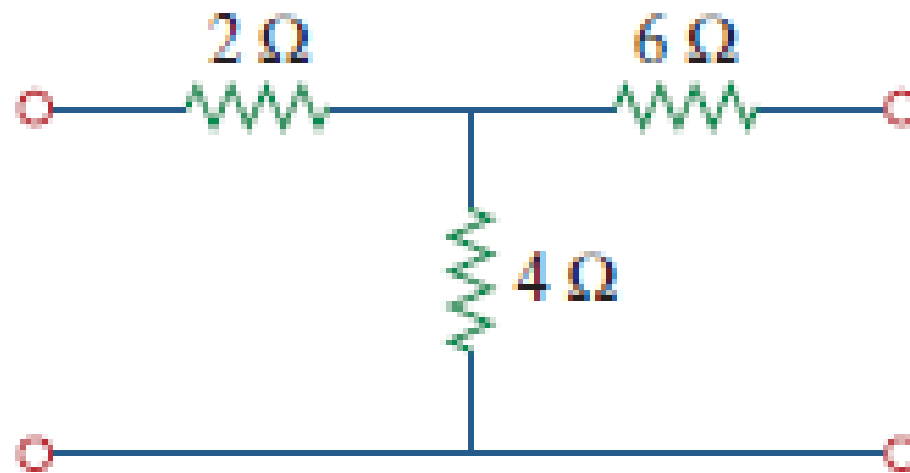
$$\mathbf{I}_1 = \frac{-\mathbf{I}_2}{6} = \left(\frac{-1}{6}\right)\left(\frac{1}{2}\mathbf{V}_2\right) = \frac{-\mathbf{V}_2}{12}$$

$$y_{12} = \frac{\mathbf{I}_1}{\mathbf{V}_2} = \frac{-1}{12} = y_{21}$$

$$\underline{[y]} = \underline{\begin{bmatrix} \frac{1}{8} & \frac{-1}{12} \\ \frac{-1}{12} & \frac{1}{2} \end{bmatrix}} \mathbf{S}$$

Numerical-2

Obtain the y parameters for the T network shown



Answer: $y_{11} = 227.3\ \text{mS}$, $y_{12} = y_{21} = -90.91\ \text{mS}$, $y_{22} = 136.36\ \text{mS}$.